

TWO-SIDED BOUNDS FOR THE MUTUAL CORRELATED AGREEMENT THRESHOLD

ABSTRACT. For Reed–Solomon codes on smooth multiplicative domains with $k \leq 2^{40}$ and $|\mathbb{F}| < 2^{256}$ — the parameter box singled out by the Proximity Prize — we bound the grand mutual correlated agreement threshold $\delta_C^*(2^{-128})$ from both sides. For $|\mathbb{F}| < 2^{128}$ the threshold is exactly 0. For $|\mathbb{F}| \geq 2^{128}$ we prove, unconditionally and by elementary self-contained arguments, that at the four challenge rates the threshold lies between one third of the relative minimum distance and $1 - \rho - 2^{-9}$ (gap 2^{-10} at rate $\frac{1}{16}$), so it never reaches capacity; a single classical import, the unique-decoding proximity gap of Ben-Sasson–Carmon–Ishai–Kopparty–Saraf, raises the lower edge to one half of the distance, determining the threshold within a factor smaller than two whenever $|\mathbb{F}| \geq 2^{128}n$. Below half the minimum distance we show that mutual and plain correlated agreement coincide up to an explicit tangent term of at most $\lfloor \delta n \rfloor$ slopes, so the remaining gap is a question about plain correlated agreement. We further prove two barriers showing that the mechanisms used here provably cannot close that gap: locator-prefix floors stop at an explicit entropy envelope, and the deep-point conversion cannot certify error beyond $\frac{1}{k}$. What remains for an exact determination is isolated as two precise open problems.

1. INTRODUCTION AND MAIN RESULT

The Proximity Prize [PP26] and the challenge collection of Arnon, Boneh and Fenzi [ABF26] ask for the largest proximity parameter δ at which mutual correlated agreement (MCA) holds for a Reed–Solomon code with error at most a stated target, and single out one parameter box: “we are mostly interested in determining δ_C^* when $L \subseteq \mathbb{F}$ is a smooth domain, $k \leq 2^{40}$, and $|\mathbb{F}| < 2^{256}$.” This paper is a self-contained two-sided answer for that box. Write $C = \text{RS}[\mathbb{F}, D, k]$ for the code of polynomials of degree less than k evaluated on a domain D of size n , $\rho = k/n$, $q = |\mathbb{F}|$, and let $\delta_C^*(\varepsilon^*)$ be the threshold of Definition 2.1 below, at the grand challenge target $\varepsilon^* = 2^{-128}$.

Theorem 1.1. *Let $D \subseteq \mathbb{F}^\times$ be a coset of a cyclic subgroup of even order n , let $C = \text{RS}[\mathbb{F}, D, k]$ with $\rho = k/n \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$, $k \leq 2^{40}$ and $n \geq 2^{15}$, and put $g := 2^{-9}$ for $\rho \geq \frac{1}{8}$ and $g := 2^{-10}$ for $\rho = \frac{1}{16}$.*

- (1) *If $q < 2^{128}$, then $\delta_C^*(2^{-128}) = 0$.*
- (2) *If $q \geq 2^{128}$, then $\delta_C^*(2^{-128}) \leq 1 - \rho - g$; in particular the threshold is bounded away from capacity by an absolute constant.*
- (3) *If moreover $q \geq 2^{128}n$, then unconditionally $\frac{1}{n} \lfloor \frac{n-k}{3} \rfloor \leq \delta_C^*(2^{-128}) \leq 1 - \rho - g$, a determination within a multiplicative factor smaller than three; and, using additionally the unique-decoding proximity gap of [BCIKS20], $\frac{1}{n} \lfloor \frac{n-k}{2} \rfloor \leq \delta_C^*(2^{-128}) \leq 1 - \rho - g$, a determination within a factor smaller than two.*
- (4) *If $2^{128} \leq q < 2^{128}n$, the same upper bound holds, and the unconditional lower bound is r^*/n with $r^* := \min(\lfloor (n-k)/3 \rfloor, \lfloor 2^{-128}q \rfloor - 1)$, which degrades to 0 as q approaches 2^{128} ; no factor claim is made in this thin regime.*

Every ingredient except the single import in (3) is proved here from first principles: a graded locator-prefix pigeonhole (Lemma 3.1), a deep-point list-to-correlated-agreement conversion (Theorem 3.2), a deep-regime theorem showing MCA *holds* with error $(\lfloor \delta n \rfloor + 1)/q$ below one third of the minimum distance for every linear code (Theorem 4.1), and a theorem showing that below one half of the distance mutual and plain correlated agreement coincide up to an explicit tangent term (Theorem 4.2). No value-set counting, exponential-sum estimate, or equidistribution input is used. The one import, [BCIKS20, Thm. 4.1], is a published theorem, so clause (3) stands on the literature as stated; we track it separately only to delimit the self-contained core. Section 6 then proves two barriers — an entropy envelope past which no prefix

floor can certify failure (Proposition 6.1), and a ceiling of $\frac{1}{k}$ on the error certifiable through the conversion (Proposition 6.2) — and isolates what remains for an exact determination as Problems 6.4 and 6.5. Remarks 3.6–3.7 complete the parameter box: characteristic-two additive smooth domains and odd 3-smooth orders satisfy the same bounds, small n down to 2^{12} retains a nonvacuous cap of gap $3/n$, and rates outside the four challenge values are covered by the same finite criterion.

The upper bounds sharpen, and the composition follows, a line of work on capacity-edge obstructions: the conditional universal cap of [Cho26b] (gap 2^{-11} under $|\mathbb{F}| \geq 2n$), the locator framework of [Cho26a], and the conversion perspective of [CS25]; the strongest known positive results for plain correlated agreement are [BCIKS20, BCHKS25], reaching the Johnson radius. A companion manuscript [Cho26b] develops the deployed-parameter refinements (subfield-pigeonhole floors widening the band to gap $\approx 2^{-4.96}$ at $n = 2^{21}$, circle and genus-one rows, explicit witnesses) and the structured form of the remaining problem; the present paper is the minimal self-contained core that answers the quoted question.

2. DEFINITIONS

Throughout, $\mathbb{F}$ is a finite field, $q = |\mathbb{F}|$, $D \subseteq \mathbb{F}$ has $|D| = n$, and $C = \text{RS}[\mathbb{F}, D, k] := \{p|_D : p \in \mathbb{F}[X], \deg p < k\}$ with $1 \leq k < n$, so C is linear of dimension k and minimum Hamming weight $n - k + 1$. For $u, v \in \mathbb{F}^D$, $\text{dist}(u, v)$ is the relative Hamming distance, and for a set $C' \subseteq \mathbb{F}^D$, $\text{dist}(u, C') := \min_{c \in C'} \text{dist}(u, c)$ and $\Lambda(C', \delta, u) := \{c \in C' : \text{dist}(u, c) \leq \delta\}$. For a pair $(f_1, f_2) \in \mathbb{F}^D \times \mathbb{F}^D$ the column distance to $C^{\equiv 2} := C \times C$ is $\text{dist}_2((f_1, f_2), C^{\equiv 2}) := \min_{c_1, c_2 \in C} \frac{1}{n} \#\{j : (f_1(j), f_2(j)) \neq (c_1(j), c_2(j))\}$. All slopes are finite: lines are $\gamma \mapsto f_1 + \gamma f_2$, $\gamma \in \mathbb{F}$.

Fix $\delta \in (0, 1)$. A slope γ is δ -CA-bad for the pair (f_1, f_2) if the point $f_1 + \gamma f_2$ is δ -close to C while the pair is δ -far in columns, $\text{dist}_2((f_1, f_2), C^{\equiv 2}) > \delta$. Following [ABF26, Defs. 4.1, 4.3], a slope γ is δ -MCA-bad for (f_1, f_2) if there exist $c \in C$ and $S \subseteq D$ with $|S| \geq (1 - \delta)n$ and $(f_1 + \gamma f_2)|_S = c|_S$, such that no pair $(c_1, c_2) \in C^{\equiv 2}$ has $f_1|_S = c_1|_S$ and $f_2|_S = c_2|_S$; that is, some witness of the point's proximity fails to extend mutually to the pair. The two errors are

$$\varepsilon_{\text{ca}}(C, \delta) := \max_{(f_1, f_2)} \frac{1}{q} \#\{\gamma \text{ } \delta\text{-CA-bad}\}, \quad \varepsilon_{\text{mca}}(C, \delta) := \max_{(f_1, f_2)} \frac{1}{q} \#\{\gamma \text{ } \delta\text{-MCA-bad}\}.$$

Definition 2.1. For a target $\varepsilon^* \in (0, 1)$, the grand MCA threshold is $\delta_C^*(\varepsilon^*) := \sup\{\delta \in (0, 1 - \rho) : \varepsilon_{\text{mca}}(C, \delta) \leq \varepsilon^*\}$, with $\sup \emptyset := 0$.

The restriction to sub-capacity radii matches the challenge convention of [ABF26, PP26]; at $\delta \geq 1 - \rho$ every word is δ -close to C by interpolation and the threshold question changes character. We record the comparison of the two errors.

Lemma 2.2. Every δ -CA-bad slope is δ -MCA-bad; hence $\varepsilon_{\text{ca}}(C, \delta) \leq \varepsilon_{\text{mca}}(C, \delta)$.

Proof. If γ is CA-bad, then $\text{dist}(f_1 + \gamma f_2, C) \leq \delta$, so a witness (c, S) with $|S| \geq (1 - \delta)n$ exists; and $\text{dist}_2((f_1, f_2), C^{\equiv 2}) > \delta$ means no set of $\geq (1 - \delta)n$ coordinates carries a pair explanation, in particular S does not. So γ is MCA-bad. $\square$

We use the standard entropy estimates for binomial coefficients, with $H_2(x) = -x \log_2 x - (1 - x) \log_2 (1 - x)$: for integers $0 < m < N$,

$$(1) \quad 2^{NH_2(m/N)} / (N + 1) \leq \binom{N}{m} \leq 2^{NH_2(m/N)}.$$

Indeed, with $p = m/N$, expanding $1 = \sum_i \binom{N}{i} p^i (1 - p)^{N-i}$ gives the upper bound by keeping the term $i = m$, which equals $\binom{N}{m} 2^{-NH_2(p)}$; and since the term ratios $\binom{N}{i+1} p^{i+1} (1 - p)^{N-i-1} / \binom{N}{i} p^i (1 - p)^{N-i} = (N - i)p / ((i + 1)(1 - p))$ exceed 1 for $i < m$ and are below 1 for $i \geq m$, the $i = m$ term is the largest of the $N + 1$ terms, giving the lower bound.

3. THE UNSAFE SIDE

The failure mechanism has two independent parts: a pigeonhole construction of a single received word carrying an enormous list at an agreement slightly above k , and a conversion of any such list into correlated-agreement failure. Neither part uses any arithmetic property of $\mathbb{F}$ beyond its size.

Lemma 3.1 (graded prefix floor). *Let $\varphi \in \mathbb{F}[X]$ have degree $c \geq 2$ and suppose the fibers of φ inside D are complete and of uniform size c : every nonempty set $\varphi^{-1}(y) \cap D$, $y \in Q := \varphi(D)$, has exactly c elements. Put $N := n/c$, let $B \subseteq \mathbb{F}$ be any subfield with $D \subseteq B$ and $\varphi \in B[X]$ (always $B = \mathbb{F}$ works), and let $1 \leq K \leq n$ and $\lceil K/c \rceil \leq m \leq N$, $w := m - \lceil K/c \rceil$. Then some B -valued word $U: D \rightarrow B$ satisfies*

$$|\Lambda(\text{RS}[\mathbb{F}, D, K], 1 - \frac{mc}{n}, U)| \geq \binom{N}{m} / |B|^w.$$

Proof. For each m -subset $M \subseteq Q$ put $\Lambda_M := \prod_{b \in M} (\varphi - b) = \sum_{j=0}^m (-1)^j e_j(M) \varphi^{m-j}$, where e_j is the j -th elementary symmetric function; $\Lambda_M \in B[X]$ has degree cm and vanishes at $x \in D$ exactly when $\varphi(x) \in M$, i.e. at exactly cm points by fiber completeness. For $z \in B^w$ let $U_z := (\varphi^m + \sum_{j=1}^w z_j \varphi^{m-j})|_D$, a B -valued word. By pigeonhole over the prefix map $M \mapsto z(M) := ((-1)^j e_j(M))_{j \leq w} \in B^w$, some z^* is attained by at least $\binom{N}{m} / |B|^w$ sets M ; for each such M set $c_M := U_{z^*} - \Lambda_M|_D = -\sum_{j>w} (-1)^j e_j(M) \varphi^{m-j}|_D$, a polynomial of degree at most $c(m-w-1) = c(\lceil K/c \rceil - 1) \leq K-1$, so $c_M \in \text{RS}[\mathbb{F}, D, K]$, and U_{z^*} agrees with c_M on the cm fiber points of M . The map $M \mapsto c_M$ is injective on the pigeonhole class: equality of words of degree $\leq K-1 \leq n-1$ on all of D forces equality of polynomials, and the powers φ^{m-j} have distinct degrees $c(m-j)$, so all $e_j(M) = e_j(M')$ for $j > w$; for $j \leq w$ they agree through z^* ; hence $\prod_{b \in M} (Y - b) = \prod_{b \in M'} (Y - b)$ and $M = M'$. $\square$

Theorem 3.2 (deep-point conversion). *Let $C = \text{RS}[\mathbb{F}, D, k]$, $q > n$, let $\delta \in (0, 1)$ satisfy $\lfloor \delta n \rfloor \leq n - k - 1$, and let $\eta \in (0, 1)$. If $\varepsilon_{\text{ca}}(C, \delta) \leq \eta \frac{1}{k} (1 - \frac{n}{q})$, then every received word $U: D \rightarrow \mathbb{F}$ satisfies $|\Lambda(\text{RS}[\mathbb{F}, D, k+1], \delta, U)| \leq q \varepsilon_{\text{ca}}(C, \delta) / (1 - \eta)$.*

Proof. Put $\varepsilon := \varepsilon_{\text{ca}}(C, \delta)$ and $a := n - \lfloor \delta n \rfloor \geq k+1$, and let $P_1, \dots, P_L \in \mathbb{F}[X]_{<k+1}$ be distinct polynomials whose restrictions lie in $\Lambda(\text{RS}[\mathbb{F}, D, k+1], \delta, U)$; each agrees with U on a set S_i of at least a points, distances being integral. Let $\Omega := \mathbb{F} \setminus D$, of size $q-n$, and for $\alpha \in \Omega$ define the simple-pole pair $f_\alpha(x) := U(x)/(x-\alpha)$, $g_\alpha(x) := -1/(x-\alpha)$ for $x \in D$. For every i and every $x \in S_i$, $f_\alpha(x) + P_i(\alpha)g_\alpha(x) = (P_i(x) - P_i(\alpha))/(x-\alpha)$, the restriction of a polynomial of degree less than k ; so the point at slope $P_i(\alpha)$ agrees with a codeword of C on S_i , hence is δ -close to C . The pair is δ -far from $C^{\equiv 2}$: if $G \in \mathbb{F}[X]_{<k}$ agreed with g_α on more than k points of D , then $(X-\alpha)G+1$, of degree at most k , would have more than k roots while taking value 1 at α ; so any pair explanation on $\geq a \geq k+1$ points is impossible. Therefore each distinct value among $P_1(\alpha), \dots, P_L(\alpha)$ is a δ -CA-bad slope, and, writing $M(\alpha)$ for the number of distinct values, $M(\alpha) \leq \varepsilon q$ for every $\alpha \in \Omega$.

It remains to pick a good α . For $i \neq j$ the nonzero polynomial $P_i - P_j$ has degree at most k , hence at most k roots, so the number of colliding pairs summed over $\alpha \in \Omega$ is at most $k \binom{L}{2}$; fix α whose collision count is at most $k \binom{L}{2} / (q-n)$. If the fibers of $i \mapsto P_i(\alpha)$ have sizes $m_1, \dots, m_{M(\alpha)}$, then $\sum_r m_r = L$ and $\sum_r m_r^2 = L + 2 \sum_r \binom{m_r}{2} \leq L + kL(L-1)/(q-n)$, so Cauchy-Schwarz gives $L^2 \leq M(\alpha) \sum_r m_r^2$, i.e. $M(\alpha) \geq L(q-n)/(q-n+k(L-1)) \geq L(q-n)/(q-n+kL)$. Combining, $\varepsilon q(q-n+kL) \geq L(q-n)$, i.e. $L(q-n)(1-\varepsilon qk/(q-n)) \leq \varepsilon q(q-n)$; the hypothesis gives $\varepsilon qk/(q-n) \leq \eta < 1$, whence $L \leq \varepsilon q/(1-\eta)$. $\square$

Theorem 3.3 (cap criterion). *In the setting of Lemma 3.1 with $K = k+1$ and $q > n$: if $\binom{N}{m} > |B|^w \binom{q}{k} + 1$, then for every $\delta \in [1 - \frac{mc}{n}, 1 - \rho)$,*

$$\varepsilon_{\text{mca}}(C, \delta) \geq \varepsilon_{\text{ca}}(C, \delta) > \frac{1}{2k} \left(1 - \frac{n}{q}\right).$$

Proof. Every $\delta < 1 - \rho$ has $\lfloor \delta n \rfloor \leq n - k - 1$, since $\delta n < n - k$ and distances are integral; so Theorem 3.2 applies at every radius in the stated range. The word U of Lemma 3.1 has $|\Lambda(\text{RS}[\mathbb{F}, D, k+1], \delta, U)| > \frac{q}{k} + 1 > \frac{q-n}{k}$ at every $\delta \geq 1 - \frac{mc}{n}$, because the $\binom{N}{m}/|B|^w$ listed codewords agree with U on $mc \geq (1 - \delta)n$ points. If $\varepsilon_{\text{ca}}(C, \delta) \leq \frac{1}{2k}(1 - \frac{n}{q})$, then Theorem 3.2 with $\eta = \frac{1}{2}$ bounds that list by $2q\varepsilon_{\text{ca}}(C, \delta) \leq \frac{q-n}{k}$, a contradiction. Lemma 2.2 transfers the bound to ε_{mca} . $\square$

Corollary 3.4 (universal cap on the challenge box). *Let $D \subseteq \mathbb{F}^\times$ be a coset of a cyclic subgroup of even order $n \geq 2^{15}$, let $\rho \in \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}\}$ and $n < q < 2^{256}$, and let g be as in Theorem 1.1. Then $\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k}(1 - \frac{n}{q})$ for every $\delta \in [1 - \rho - g, 1 - \rho]$.*

Proof. Since n is even and $n \mid q-1$, the field has odd characteristic and -1 is the element of order 2 in the subgroup H with $D = \alpha H$; so $\varphi := X^2$ has fibers $\{x, -x\}$ inside D , complete and of size 2. Apply Lemma 3.1 and Theorem 3.3 with $c = 2$, $B = \mathbb{F}$, $K = k+1$ and $m := \lceil (k+1+gn)/2 \rceil$; then $2m \geq k+gn$, so $[1 - \rho - g, 1 - \rho] \subseteq [1 - \frac{2m}{n}, 1 - \rho]$, and $w = m - \lceil (k+1)/2 \rceil \leq (gn+2)/2$, and $m \leq N = n/2$ since $k+gn+2 \leq n$. It remains to verify the criterion. Writing $\beta := \log_2 q \leq 256$, by (1) it suffices that

$$(2) \quad \frac{n}{2} H_2(m/N) \geq 128gn + 512 + \log_2\left(\frac{n}{2} + 1\right),$$

since $\beta w \leq 128gn + 256$ and $\log_2(\frac{q}{k} + 1) \leq 256$. Here $m/N = 2m/n \in [\rho + g, \rho + g + 2^{-13}]$ for $n \geq 2^{15}$, and on that interval the concave H_2 is at least $h(\rho) := \min(H_2(\rho+g), H_2(\rho+g+2^{-13}))$, which evaluates (rounding down) to $h \geq 0.9999, 0.8122, 0.5489, 0.3410$ at $\rho = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$. Both sides of (2) are linear in n up to the logarithm, with slopes $h/2$ and $128g$, and $h/2 - 128g$ equals $0.2499, 0.1561, 0.0244, 0.0455$ respectively, all positive; so it suffices to check (2) at $n = 2^{15}$, where the left side is at least $16384h \geq 8993$ (the tightest case, $\rho = \frac{1}{8}$) against a right side of $8192 + 512 + 15 = 8719$ there, and 5586 against $4096 + 527$ at $\rho = \frac{1}{16}$, and larger margins at $\rho \in \{\frac{1}{4}, \frac{1}{2}\}$; the slack of at least 274 bits makes the criterion of Theorem 3.3 strict. (All four instances were additionally verified by exact integer arithmetic at $n = 2^{15}$, $q = 2^{256} - 1$.) $\square$

Proposition 3.5 (small fields). *For every linear code $C \subsetneq \mathbb{F}^n$ and every $\delta \in (0, 1)$, $\varepsilon_{\text{mca}}(C, \delta) \geq 1/q$. Hence $\delta_C^*(\varepsilon^*) = 0$ whenever $\varepsilon^* < 1/q$, in particular whenever $q < 2^{128}$ at the grand target. At the boundary $q = 2^{128}$ the same construction gives $\varepsilon_{\text{mca}} \geq 2^{-128}$ at every radius, which meets the threshold condition with equality and does not by itself force degeneracy.*

Proof. Pick $f_2 \notin C$, any $c_0 \in C$, $\gamma_0 \in \mathbb{F}$, and set $f_1 := c_0 - \gamma_0 f_2$. Then $f_1 + \gamma_0 f_2 = c_0$, so (c_0, D) is a proximity witness with $S = D$; but a pair explanation on $S = D$ would force $f_1, f_2 \in C$, which fails. Hence γ_0 is δ -MCA-bad for this pair at every δ , and $\varepsilon_{\text{mca}}(C, \delta) \geq 1/q$. $\square$

Remark 3.6 (other smooth domains). Lemma 3.1 and Theorem 3.3 are stated for arbitrary complete-fiber polynomial maps, so Corollary 3.4 extends beyond even-order multiplicative cosets with the identical arithmetic. (i) *Characteristic two, additive smoothness.* If $D = t + V$ is an $\mathbb{F}_2$ -affine subspace of dimension $s \geq 1$ and $a \in V \setminus \{0\}$, the linearized polynomial $\varphi(x) = x^2 + ax$ satisfies $\varphi(x+a) = \varphi(x)$, so its fibers inside D are the pairs $\{x, x+a\}$, complete and of size 2; the proof of Corollary 3.4 then goes through verbatim. (ii) *Odd smooth orders.* If $3 \mid n$ (e.g. n a power of 3), take $\varphi = X^3$: the fibers are the μ_3 -cosets, complete of size 3 since $\mu_3 \subseteq H$ and $3 \mid q-1$ forces characteristic $\neq 3$; the same entropy verification with $N = n/3$ yields the four gaps of Corollary 3.4 for $n \geq 2^{16}$ (checked as in the proof above, tightest margin again at $\rho = \frac{1}{8}$). (iii) Chebyshev and circle-code domains, and genus-one (ECFFT) domains, also carry complete uniform fibers and receive the same caps; see [Cho26b].

Remark 3.7 (small n and general rates). For $2^{12} \leq n < 2^{15}$ the corollary's uniform gap is not claimed, but capacity is still refuted: taking $m = \lceil (k+1)/2 \rceil + 1$, so $w = 1$ and agreement $2m \geq k+3$, the criterion $\binom{n/2}{m} > q(\frac{q}{k} + 1)$ holds at all four rates for $n \geq 2^{12}$ (at $\rho = \frac{1}{16}$, $n = 2^{12}$: $\log_2 \binom{2048}{130} \geq 687$ against at most 505), giving $\delta_C^*(2^{-128}) \leq 1 - \rho - 3/n$ whenever $q \geq 2^{128}$. For $n < 2^{12}$ with $q \geq 2^{128}$ our method gives nothing above the safe region, and we record this corner as genuinely open. For rates outside the four challenge values, Theorem 3.3 applies unchanged;

the achievable gap is governed by the criterion, asymptotically the envelope of Proposition 6.1, and is positive for every fixed $\rho \in (0, 1)$ and large n . Finally, the list challenge of [ABF26] is unsafe at the same first step: Lemma 3.1 at $K = k$ produces a single word with more than $2^{-128}q$ listed codewords under the analogous criterion, with the same verification.

4. THE SAFE SIDE

We now prove that MCA *holds*, with essentially optimal error, deep inside the unique-decoding region — for every linear code — and that up to half the minimum distance the mutual layer costs only an explicit tangent term.

Theorem 4.1 (deep regime). *Let $C \subseteq \mathbb{F}^n$ be linear with minimum weight $w_{\min}$, let $\delta \in (0, 1)$, $r := \lfloor \delta n \rfloor$, and assume $3r \leq w_{\min} - 1$. Then for every pair (f_1, f_2) at least one of the following holds:*

- (1) *there are $c_1, c_2 \in C$ and $T \subseteq [n]$ with $|T| \leq r$ and $f_i = c_i$ off T for $i = 1, 2$; or*
- (2) *at most $r + 1$ slopes γ have $\text{dist}(f_1 + \gamma f_2, C) \leq \delta$.*

Consequently $\varepsilon_{\text{ca}}(C, \delta) \leq (r + 1)/q$ and $\varepsilon_{\text{mca}}(C, \delta) \leq (r + 1)/q$.

Proof. Suppose (2) fails, so there are $r + 2$ distinct slopes $\gamma_0, \dots, \gamma_{r+1}$, codewords c_j , and error sets $E_j := \{i : (f_1 + \gamma_j f_2)(i) \neq c_j(i)\}$ with $|E_j| \leq r$. For $j \neq l$, solving the two agreements off $E_j \cup E_l$ gives $f_2 = d_{jl} := (c_j - c_l)/(\gamma_j - \gamma_l)$ and $f_1 = e_{jl} := (\gamma_j c_l - \gamma_l c_j)/(\gamma_j - \gamma_l)$ off $E_j \cup E_l$, with $d_{jl}, e_{jl} \in C$. If $r = 0$ all E_j are empty and (1) holds with $T = \emptyset$; so assume $r \geq 1$, hence at least three slopes. For any three indices j, l, m , the codeword $d_{jl} - d_{jm}$ vanishes off $E_j \cup E_l \cup E_m$, a set of size at most $3r \leq w_{\min} - 1$, so $d_{jl} = d_{jm}$; chaining through shared indices, all d_{jl} equal a single $d \in C$, and likewise all e_{jl} equal a single $e \in C$. Let $T := \{i : (f_1(i), f_2(i)) \neq (e(i), d(i))\}$. If $i \notin E_j$ and $i \notin E_l$ for two distinct indices, then solving at coordinate i gives $f_2(i) = d(i)$ and $f_1(i) = e(i)$, so $i \notin T$; contrapositively each $i \in T$ lies in at least $r + 1$ of the $r + 2$ sets E_j , whence $(r + 1)|T| \leq \sum_j |E_j| \leq r(r + 2)$ and $|T| \leq r(r + 2)/(r + 1) < r + 1$, i.e. $|T| \leq r$. This is (1) with $(c_1, c_2) = (e, d)$.

For the error bounds, fix a pair. In case (2), both CA-bad and MCA-bad slopes are among the at most $r + 1$ close slopes. In case (1): the pair is within column distance $r/n \leq \delta$ of (c_1, c_2) , so no slope is CA-bad. For MCA, write $\epsilon_i := f_i - c_i$, supported on T , and call γ *tangent* if $\epsilon_1(i) + \gamma \epsilon_2(i) = 0$ for some $i \in T$; there are at most $|T| \leq r$ tangent slopes, since a coordinate with $\epsilon_2(i) \neq 0$ determines one value of γ and a coordinate with $\epsilon_2(i) = 0 \neq \epsilon_1(i)$ determines none. Let γ be non-tangent and let (c, S) be any proximity witness for $f_1 + \gamma f_2$, so $|S| \geq n - r$. The codeword $c - (c_1 + \gamma c_2)$ vanishes on $S \setminus T$, of size at least $n - 2r$, so its weight is at most $2r \leq w_{\min} - 1$ and it is zero; then $(\epsilon_1 + \gamma \epsilon_2)|_S = 0$, and since γ is non-tangent this function is nonzero at every point of T , forcing $S \cap T = \emptyset$ and hence $f_i|_S = c_i|_S$: the witness extends mutually. So every MCA-bad slope is tangent, and there are at most $r \leq r + 1$ of them. $\square$

Theorem 4.2 (mutual from correlated, below half the distance). *Let $C \subseteq \mathbb{F}^n$ be linear with minimum weight $w_{\min}$, $\delta \in (0, 1)$, $r := \lfloor \delta n \rfloor$, and assume $2r \leq w_{\min} - 1$. Then $\varepsilon_{\text{mca}}(C, \delta) \leq \max(\varepsilon_{\text{ca}}(C, \delta), r/q)$.*

Proof. Fix a pair (f_1, f_2) . If $\text{dist}_2((f_1, f_2), C^{\equiv 2}) > \delta$, every MCA-bad slope is CA-bad: proximity gives a witness, fairness denies every pair explanation of size $\geq (1 - \delta)n$, and the CA conditions are exactly these two; so the count is at most $q \varepsilon_{\text{ca}}(C, \delta)$. If instead $\text{dist}_2 \leq \delta$, fix an explanation $(p_1, p_2) \in C^{\equiv 2}$ with column error set E , $|E| \leq r$, and put $\epsilon_i := f_i - p_i$, supported on E . Exactly as in the proof of Theorem 4.1 — with $2r \leq w_{\min} - 1$ now doing the forcing — for every non-tangent γ and every witness (c, S) , the codeword $c - (p_1 + \gamma p_2)$ vanishes on $S \setminus E$ of size $\geq n - 2r$, hence is zero; then S avoids the support of $\epsilon_1 + \gamma \epsilon_2$, which is all of E , so $f_i|_S = p_i|_S$ and the witness extends. MCA-bad slopes are thus among the at most $|E| \leq r$ tangent values. $\square$

Corollary 4.3 (safe to half the distance, modulo one import). *Let $C = \text{RS}[\mathbb{F}, D, k]$ and $\delta \leq \frac{n-k}{2n}$. Assuming the unique-decoding proximity gap of [BCIKS20, Thm. 4.1], namely $\varepsilon_{\text{ca}}(C, \delta) \leq n/q$ for δ up to half the minimum distance, we get $\varepsilon_{\text{mca}}(C, \delta) \leq n/q$.*

Proof. Here $w_{\min} = n - k + 1$ and $2\lfloor \delta n \rfloor \leq n - k = w_{\min} - 1$, so Theorem 4.2 gives $\varepsilon_{\text{mca}} \leq \max(n/q, \lfloor \delta n \rfloor / q) = n/q$. $\square$

5. PROOF OF THEOREM 1.1

(1) is Proposition 3.5: $\varepsilon_{\text{mca}}(C, \delta) \geq 1/q > 2^{-128}$ at every δ , so the supremum set is empty.

(2) By Corollary 3.4, $\varepsilon_{\text{mca}}(C, \delta) > \frac{1}{2k}(1 - \frac{n}{q})$ for every $\delta \in [1 - \rho - g, 1 - \rho]$. Here $k \leq 2^{40}$ gives $\frac{1}{2k} \geq 2^{-41}$, and $n \leq 16k \leq 2^{44}$ with $q \geq 2^{128}$ gives $1 - \frac{n}{q} \geq \frac{1}{2}$; so $\varepsilon_{\text{mca}}(C, \delta) > 2^{-42} > 2^{-128}$ throughout the band, no radius there belongs to the supremum set, every member of that set is below $1 - \rho - g$, and $\delta_C^*(2^{-128}) \leq 1 - \rho - g$.

(3) and (4). Let $r^* := \min(\lfloor (n - k)/3 \rfloor, \lfloor 2^{-128}q \rfloor - 1)$ and suppose $r^* \geq 1$. At $\delta := r^*/n$ we have $\lfloor \delta n \rfloor = r^*$ and $3r^* \leq n - k = w_{\min} - 1$, so Theorem 4.1 gives $\varepsilon_{\text{mca}}(C, \delta) \leq (r^* + 1)/q \leq 2^{-128}$, using $r^* + 1 \leq \lfloor 2^{-128}q \rfloor$; also $\delta < 1 - \rho$, so δ lies in the supremum set and $\delta^* \geq r^*/n$. If $q \geq 2^{128}n$ then $\lfloor 2^{-128}q \rfloor - 1 \geq n - 1 \geq \lfloor (n - k)/3 \rfloor$, so $r^* = \lfloor (n - k)/3 \rfloor$ and $\delta^* \geq \frac{1}{n} \lfloor (n - k)/3 \rfloor \geq \frac{1 - \rho}{3} - \frac{2}{3n}$; against the upper bound $1 - \rho - g$ the ratio is at most 2.997 at the four rates for $n \geq 2^{15}$, a factor smaller than three. With the import, take $\delta := \frac{1}{n} \lfloor (n - k)/2 \rfloor \leq \frac{n - k}{2n}$: Corollary 4.3 gives $\varepsilon_{\text{mca}} \leq n/q \leq 2^{-128}$ when $q \geq 2^{128}n$, so $\delta^* \geq \frac{1 - \rho}{2} - \frac{1}{2n}$, and the ratio to the upper bound is at most 1.998 at the four rates, a factor smaller than two. In the thin regime $2^{128} \leq q < 2^{128}n$ only the r^* clause is claimed; as $q \downarrow 2^{128}$, r^* reaches 0 and the lower bound becomes vacuous, as stated. $\square$

Two comments on the statement. First, the import in (3) is a published theorem, so the two-sided determination within a factor of two stands on the literature; the unconditional clause records what the present paper proves alone. Second, the case split at $q = 2^{128}$ is genuinely necessary at the stated target: below it the threshold degenerates, at it neither degeneracy nor a nontrivial safe radius follows (Proposition 3.5), and the meaningful question there is posed at a larger target, as the deployed circle-code rows already illustrate [Cho26b].

6. BARRIERS, AND WHAT REMAINS

The gap between the two edges of Theorem 1.1 is not an artifact of lazy constants: each mechanism above provably stops where it stops.

Proposition 6.1 (entropy envelope for prefix floors). *In the setting of Lemma 3.1, write $\beta := \log_2 |B|$, $A := mc$ and $g := (A - k)/n$, and let $K \leq k + 1$. If $H_2(\rho + g) \leq \beta(g - \frac{c}{n})$, then $\binom{N}{m} \leq |B|^w$: the pigeonhole class is not guaranteed to exceed a single codeword, and no certificate of the form of Theorem 3.3 exists at agreement A and scale c . In particular, over any field with no proper subfield containing D and $q \geq 2^{128}$ (so $\beta \geq 128$), a certificate at scale c requires $g < \frac{1}{128} + \frac{c}{n}$; since a useful certificate also needs $\binom{N}{m} > \frac{q}{k} \geq 2^{88}$, hence $N \geq 89$ and $c \leq n/89$, every certified gap is below $\frac{1}{128} + \frac{1}{89} < 2^{-5.6}$, and at the small scales used above, below $\frac{1}{128} + \frac{2}{n} \approx 2^{-7}$. For deployed rows whose domains lie in small subfields ($\beta = 31$) the same envelope, now $H_2(\rho + g^*) = g^*\beta$, is wider, and it is reached and saturated in [Cho26b].*

Proof. Here $w = m - \lceil K/c \rceil \geq m - (k + c)/c = (A - k - c)/c$, so by (1), $\log_2 \binom{N}{m} \leq \frac{n}{c} H_2(A/n) \leq \frac{n}{c} \cdot \beta(g - \frac{c}{n}) = \beta \frac{gn - c}{c} \leq \beta w$. The instantiation takes $\beta = \log_2 q \geq 128$ and $H_2 \leq 1$. $\square$

Proposition 6.2 (ceiling of the conversion). *Let δ be as in Theorem 3.2 and $T := \frac{1}{k}(1 - \frac{n}{q})$. If some word u has $|\Lambda(\text{RS}[\mathbb{F}, D, k + 1], \delta, u)| \geq 1 + \kappa q T$ with $\kappa > 0$, then $\varepsilon_{\text{ca}}(C, \delta) \geq \frac{\kappa}{\kappa + 1} T$. Conversely, no list bound, however large, certifies $\varepsilon_{\text{ca}} \geq T$ through Theorem 3.2: the certified error tends to T from below as $\kappa \rightarrow \infty$.*

Proof. Put $x := \varepsilon_{\text{ca}}(C, \delta)$; if $x \geq T$ there is nothing to prove, so let $x < T$ and $\eta := x/T < 1$. Theorem 3.2 applies with this η and yields $\kappa q T \leq q x / (1 - x/T)$, so $\kappa(T - x) = \kappa T(1 - x/T) \leq x$, i.e. $x \geq \kappa T / (\kappa + 1)$. The converse direction is read off the same inequality. $\square$

Remark 6.3 (why half the distance is a wall). The forcing step in Theorems 4.1 and 4.2 — a codeword vanishing outside a $2r$ -set must be zero — fails precisely when $2r \geq w_{\min}$, and for

$C = \text{RS}[\mathbb{F}, D, k]$ it fails with room to spare: the codewords supported inside a fixed set W with $|W| = 2r$ are the evaluations of polynomials of degree $< k$ vanishing on the $n - 2r$ points off W , a space of dimension exactly $\max(0, k - n + 2r) = \max(0, 2r - (n - k))$. Just past half the distance this ambiguity space has positive dimension and $q^{2r-(n-k)}$ elements, so witnesses of proximity are no longer forced onto the explanation line and the tangent count breaks down structurally, not numerically.

By Theorem 4.2, below half the distance the mutual and plain problems coincide up to $r/q \leq 2^{-128}$ error at the relevant sizes; so, at the four challenge rates and $q \geq 2^{128}n$, the entire remaining content of the committee's question is the following pair of problems, on the interval between the two edges of Theorem 1.1(3).

Problem 6.4 (plain correlated agreement in the band). For $C = \text{RS}[\mathbb{F}, D, k]$ as in Theorem 1.1 with $q \geq 2^{128}n$, determine whether $\varepsilon_{\text{ca}}(C, \delta) \leq 2^{-128}$ for $\delta \in (\frac{1-\rho}{2}, 1 - \rho - g)$ — equivalently, by Theorem 4.2, whether $\varepsilon_{\text{mca}} \leq 2^{-128}$ there. The known positive results [BCIKS20, BCHKS25] reach the Johnson radius $1 - \sqrt{\rho}$; the subinterval $(1 - \sqrt{\rho}, 1 - \rho - g)$ is open even for the list-size question, and Propositions 6.1 and 6.2 show the present floors and conversion cannot decide it.

Problem 6.5 (the mutual layer above half the distance). Determine $\varepsilon_{\text{mca}}(C, \delta) - \varepsilon_{\text{ca}}(C, \delta)$ for $\delta \in (\frac{1-\rho}{2}, 1 - \rho - g)$: bound the number of MCA-bad slopes carried by proximity witnesses that do not extend mutually, in the regime where the ambiguity space of Remark 6.3 is nontrivial. A polynomial bound here, combined with a resolution of Problem 6.4 up to the same radius, would replace both edges of Theorem 1.1(3) by a single threshold value.

In the companion framework [Cho26b], Problem 6.5 is organized by a stabilizer partition of witness supports over the divisor lattice of D : periodically witnessed bad slopes are counted exactly, and what remains is a single conjecture — polynomially many aperiodically witnessed bad slopes in the band — whose truth would let a finite scanner output $\delta_C^*(2^{-128})$ exactly, at a value essentially equal to the unsafe edge produced by the optimized floors (≈ 0.4679 for the deployed rate- $\frac{1}{2}$ rows). Whether by that route or another, Problems 6.4 and 6.5 are, after this paper, precisely what stands between the box singled out by the committee and its exact answer.

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